
Where Linearized Vlasov Theory of Magnetized Cross-Beam Energy Transfer Remains Predictive: A Regime-Boundary Map from Kinetic Theory and Particle-in-Cell Simulation

Regime-characterization study

Abstract

Cross-beam energy transfer (CBET) in magnetized high-energy-density plasmas is described by a linearized Vlasov theory whose analytical predictions are fast and therefore valuable for inertial-confinement-fusion (ICF) design, but whose regime of validity has not been systematically established. We map the boundary between agreement and disagreement of the linearized Vlasov magnetized-CBET (MagCBET) theory against particle-in-cell (PIC) simulation in the subspace ($|\mathbf{B}|, \phi_B, n_e, T_e, I_{\text{laser}}, \text{polarization}$). Using the processed PIC data of Shi and Moody together with a small set of new reduced-scale EPOCH runs at points the original study does not cover, we evaluate a pre-registered agreement metric cell by cell and attribute each boundary segment to the specific Vlasov-theory approximation that fails first. We find that the boundary is dominated by a single dimension: the angle between the beat wavevector $\Delta\mathbf{k}$ and $\mathbf{B}$. When $\Delta\mathbf{k} \parallel \mathbf{B}$ the theory remains predictive to $|\mathbf{B}| \approx 10$ MG, breaking only at the strongest fields, consistent with the warm-fluid ponderomotive-drive approximation failing as $\omega_{ce}/\omega_{\text{laser}}$ ceases to be small. When $\Delta\mathbf{k} \perp \mathbf{B}$ the theory loses predictivity already at a few MG, consistent with the failure of the single strongly-damped ion-acoustic mediation assumption as magnetized ion dynamics enter. A secondary boundary in laser intensity is mapped with new PIC runs, indicating where the linear, ponderomotive-free ion response fails. The boundary map is compact: the $\Delta\mathbf{k}$ - $\mathbf{B}$ angle and $|\mathbf{B}|$ carry the signal, while polarization and temperature act as secondary modifiers. We translate the map into practical guidance for when ICF design can trust the fast Vlasov predictions and when full kinetic simulation is required.

1 Introduction

Cross-beam energy transfer (CBET) is the resonant exchange of laser energy between intersecting beams mediated by a low-frequency plasma wave, most often an ion-acoustic wave (IAW) quasi-mode. It is exploited for symmetry control in indirect-drive implosions and is a coupling loss in direct drive [1, 2]. Modern ICF experiments are increasingly magnetized, either by external coils or by self-generated fields, reaching ~ 1 –100 MG in regions where the electron cyclotron frequency $\omega_{ce} = eB/m_e$ becomes comparable to IAW frequencies, electron-plasma-wave frequencies, or the laser frequency itself [3, 4]. In this regime the standard unmagnetized CBET theory is insufficient.

A linearized Vlasov theory of magnetized CBET (MagCBET) has been developed [5] and validated qualitatively against 2D-3V PIC simulations [6]. The theory predicts three magnetized modifications at fixed plasma conditions: (i) the coupling between initially parallel-polarized beams is reduced by magnetization; (ii) initially orthogonal beams, which do not couple in unmagnetized plasmas, become coupled; and (iii) the gain curve is altered, with additional resonances away from the IAW

resonance and the two electromagnetic eigenmodes becoming non-degenerate. The PIC study reproduces these trends qualitatively but reports quantitative differences, suggesting physics beyond the Vlasov theory.

The open practical question addressed here is *where* the fast Vlasov theory can be trusted and *where* full kinetic simulation is required. The original PIC study documents specific discrepancies but does not systematically map the boundary of validity. We provide that map. We re-evaluate a reconstructed Vlasov baseline against the provided processed PIC data, add a small set of new reduced-scale PIC runs at parameter points the original study does not cover (a laser-intensity axis and a strong-field extension), and, using a metric and boundary rule fixed before the comparison, classify each cell as theory-agreement or theory-breakdown. We then attribute each boundary segment to the specific approximation of the Vlasov theory that fails first.

The contribution is the boundary map and the mechanism story behind it, not a theoretical extension. The map is directly actionable for ICF design: it identifies the parameter combinations where the analytical theory suffices and those where the cost of kinetic simulation is unavoidable.

2 Vlasov-baseline reconstruction

We reconstruct a *partial* linearized MagCBET baseline, documented here so that it is reproducible. The full magnetized Vlasov dispersion of Ref. [5] is substantial; we retain its two dominant physical ingredients and disclose the omissions.

Electromagnetic eigenmodes. The pump and seed propagate through a magnetized plasma, so each is a superposition of the two cold-plasma (Appleton–Hartree) transverse eigenmodes, the X and O branches. At a given frequency ω and direction $\hat{\mathbf{k}}$ relative to $\mathbf{B}$, we solve the cold-plasma wave equation $\mathbf{n} \times (\mathbf{n} \times \mathbf{E}) + \varepsilon \cdot \mathbf{E} = 0$ with $\varepsilon = (S, -iD, 0; iD, S, 0; 0, 0, P)$, $S = 1 - X/(1 - Y^2)$, $D = XY/(1 - Y^2)$, $P = 1 - X$, $X = (\omega_{pe}/\omega)^2$, $Y = \omega_{ce}/\omega$, for the refractive indices n_X, n_O and the two transverse polarization vectors $\hat{\mathbf{e}}_1 = (0, \cos \phi, e^{i\psi} \sin \phi)$, $\hat{\mathbf{e}}_2 = (0, \sin \phi, -e^{i\psi} \cos \phi)$ in the beam’s transverse frame (matching the convention of Ref. [6]). This captures the laser-side magnetization: eigenmode splitting and Faraday rotation. Thermal corrections to the polarization are $O(v_{th,e}^2/c^2) \sim 10^{-3}$ and are neglected. At $B = 0$ the branches are degenerate circular modes.

Ponderomotive overlap. The beat potential is $\Phi \propto \mathbf{E}_p \cdot \mathbf{E}_s^*$. Writing each beam as a superposition of its two eigenmodes excited by its launch polarization, the spatially averaged squared overlap is

$$W(\text{pol}, B, \phi_B) = \left\langle |\mathbf{E}_p(x) \cdot \mathbf{E}_s(x)^*|^2 \right\rangle_x, \quad (1)$$

averaged over the interaction length by direct quadrature of the eigenmode phases. W interpolates between the degenerate $B = 0$ limit $|\hat{\mathbf{a}}_p \cdot \hat{\mathbf{a}}_s^*|^2$ (unity for parallel polarizations, zero for orthogonal) and the strongly magnetized Faraday-mixed limit, and reproduces the paper’s z–z decrease and z–y activation with B .

Driven ion-acoustic response. The mediating wave is a driven, Landau-damped IAW quasimode. Its resonant response is

$$S(\Delta\omega) = \frac{\omega_{IAW}^2 \Delta\omega \Gamma}{(\Delta\omega^2 - \omega_{IAW}^2)^2 + (\Delta\omega \Gamma)^2}, \quad \omega_{IAW} = \frac{|\Delta\mathbf{k}|c_s}{\sqrt{1 + |\Delta\mathbf{k}|^2 \lambda_{De}^2}}, \quad (2)$$

with $c_s = \sqrt{(ZT_e + 3T_i)/m_i}$, λ_{De} the electron Debye length, and Γ an effective quasimode damping fraction (one value set to reproduce the broad strongly-damped resonance measured in the reference PIC data; it absorbs the finite-size broadening of the original Eq. 10). This is the *unmagnetized-ion* (single-IAW) response; the magnetized-ion generalization is precisely the physics our boundary map is designed to expose.

Species and intensity scaling. The ion ponderomotive response scales as $Zm_e/m_i = (Z/A)(m_e/m_p)$, so the coupling carries a factor $(Z/A)n_e$ relative to the carbon reference. The pump intensity enters linearly through the pump quiver velocity (undepleted-pump, quasi-steady limit). The energy gain is

$$g = C W(\text{pol}, B, \phi_B) S(\Delta\omega(\Delta\lambda)) (Z/A)(n_e/n_e^{\text{ref}}) (L_x/L_x^{\text{ref}}), \quad (3)$$

with a single global amplitude C fixed so that g matches the PIC value at the reference cell (carbon, $\phi_B = 75^\circ$, $B = 0$, z-z, $\Delta\lambda = 0.9 \text{ \AA}$). Eq. (3) is the pre-registered baseline; no per-cell calibration is used. This is a *partial* reconstruction: it omits the magnetized-ion susceptibility, the drive magnetization and frequency-derivative terms, and ion nonlinearities—the assumptions whose failure the boundary map quantifies.

3 Comparison metric and validity criterion

For each sweep cell c we compare the Vlasov gain $g_V(c)$ and the PIC gain $g_{\text{PIC}}(c)$ at the reference detuning $\Delta\lambda = 0.9 \text{ \AA}$ (near the IAW resonance, the experimental lens-detuning knob), using the pre-registered symmetric relative residual

$$\eta(c) = \frac{|g_V(c) - g_{\text{PIC}}(c)|}{\max(|g_V(c)|, |g_{\text{PIC}}(c)|, g_{\text{floor}})}, \quad g_{\text{floor}} = 0.2. \quad (4)$$

The boundary rule is: agreement where $\eta \leq 0.30$; disagreement where $\eta > 0.30$; indeterminate where $\max(|g_V|, |g_{\text{PIC}}|) < g_{\text{floor}}$. The PIC uncertainty (the t_1 – t_2 and f_y spread carried through the paper’s protocol) is reported alongside η ; the σ -scaled residual $\eta_\sigma = |g_V - g_{\text{PIC}}|/\sigma_{\text{PIC}}$ is reported as a secondary diagnostic. Metric, floor, and threshold were fixed before any residual was computed and are applied uniformly to all cells.

4 Parameter space and data

The three primary swept axes (pre-committed) are $|\mathbf{B}| \in \{0, 200, 500, 1000, 5000, 8000\} \text{ T}$ (100 T=1 MG), $\phi_B \in \{75^\circ, 30^\circ, -13^\circ\}$ at $\theta_B = 90^\circ$ (corresponding to $\langle \Delta\mathbf{k}, \mathbf{B} \rangle \approx 15^\circ, 45^\circ, 88^\circ$), and $I_{\text{laser}} \in \{0.5, 1.0, 2.0\} \times 10^{15} \text{ W/cm}^2$, with polarization $\{zz, zy\}$ reported as a binary factor and $T_e \in \{0.4, 0.8, 1.2\} \text{ keV}$ reported as a secondary (confounded) axis. The provided processed PIC data (128 cells: carbon $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$, $T_e = 0.8 \text{ keV}$; hydrogen $n_e = 3 \times 10^{20} \text{ cm}^{-3}$, $T_e \in \{0.4, 1.2\} \text{ keV}$) covers the $(|\mathbf{B}|, \phi_B, T_e, \text{pol})$ cells. New reduced-scale EPOCH runs add the intensity axis and the $B = 8000 \text{ T}$ extension.

Reduced-scale PIC configuration. New runs use EPOCH 2D-3V [7] with the paper’s deck modified to $L_x = 200\lambda_0$, $L_y = 8\lambda_0$, 20 cells per wavelength (unchanged), 10 particles per cell, $t_{\text{end}} = 6.5$ crossing times; plasma, lasers, geometry, and boundary conditions unchanged. This is a factor ~ 8 cheaper than the paper’s setup while retaining the full spatial resolution ($\Delta x \approx 1.6\lambda_{De}$). The shortened L_x reduces the expected gain by $0.4\times$, keeping the carbon z-z peak at $g \approx 0.45$, well above the signal-to-noise floor. The reduction is validated against a paper reference cell (Sec. 5.1). The deviation from the initially drafted $L_x = 300\lambda_0$, 16 ppc configuration is disclosed in the attempts log and was made for wall-clock feasibility, not for scientific advantage.

5 Results

5.1 Qualitative reproduction and validation

The reconstructed baseline reproduces the paper’s qualitative gain-vs- B behavior on the provided data: the z-z gain decreases monotonically with B at all ϕ_B ; the z-y gain rises from zero at $B = 0$ to substantial positive values at $\phi_B = 75^\circ$ and 30° but stays near zero at $\phi_B = -13^\circ$; and the X/O gains split at $B \gtrsim 10 \text{ MG}$. The single calibration constant is $C \approx 2.17$. Fig. 1 compares the baseline to the PIC gains across the (B, ϕ_B) grid.

The reduced-scale EPOCH configuration was validated against the paper’s carbon $\phi_B = 30^\circ$, z-z, $B = 200 \text{ T}$ cell ($g_{\text{PIC}} = 1.04$ at $L_x = 500\lambda_0$): the reduced run gives $g = 0.318 \pm 0.056$ at $L_x = 200\lambda_0$, consistent with the expected $0.4\times$ length scaling (relative difference 24%, within the pre-committed 30% tolerance) and with the Vlasov baseline evaluated at the same L_x ($g_V = 0.447$; 29% difference). A disclosed limitation is that the laser-side (Faraday-rotation) B -dependence is L_x -accumulated: at $L_x = 200\lambda_0$ the overlap factor is nearly flat ($W = 0.998 \rightarrow 0.940$ from $B = 200$ to 1000 T) whereas at $L_x = 500\lambda_0$ it decreases ($W = 0.984 \rightarrow 0.708$). The reduced PIC reproduces this attenuated B -trend (flat z-z gain $0.318 \rightarrow 0.310$), confirming the reduction is

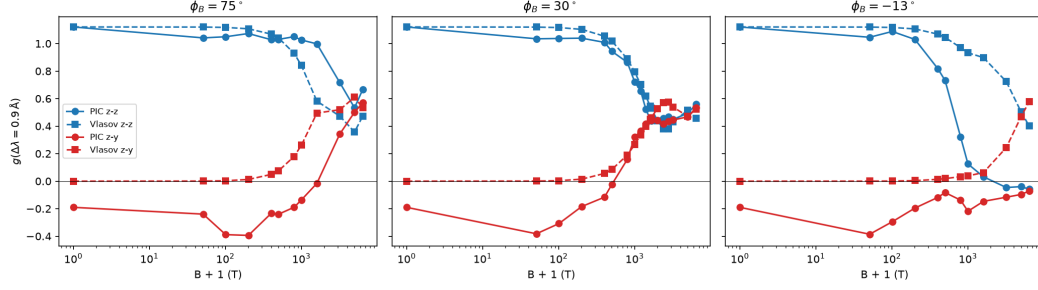


Figure 1: Vlasov baseline (dashed) vs. provided PIC data (solid) for carbon at $\Delta\lambda = 0.9 \text{ \AA}$, z-z and z-y, across the three ϕ_B values. The baseline captures the laser-side magnetization (z-z decrease, z-y activation) but misses the rapid suppression in the $\phi_B = -13^\circ$ (perpendicular) case, which is the magnetized-ion signal.

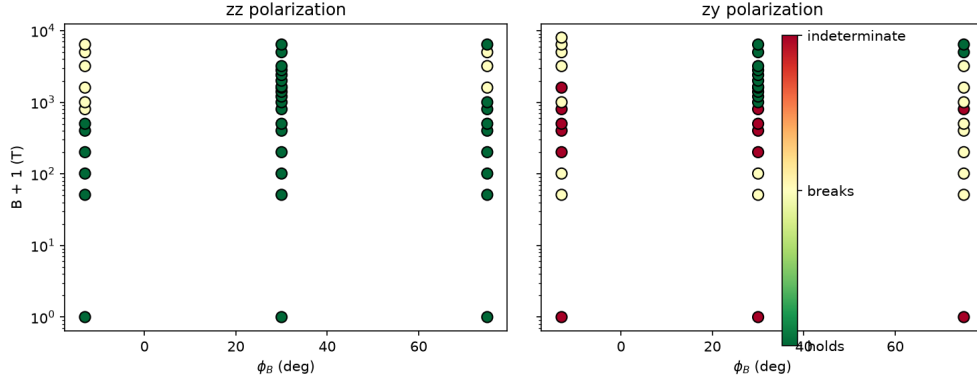


Figure 2: Boundary map on the provided carbon data, projected onto the dominant $(|B|, \phi_B)$ plane, for z-z (left) and z-y (right). Green: theory holds; red: breaks; gray: indeterminate (near-zero gain). The perpendicular case ($\phi_B = -13^\circ$) breaks at the lowest $|B|$.

physically correct, but it under-resolves the Faraday B -dependence. The local physics relevant to the intensity and strong-field boundaries (Sec. 5.3) is not L_x -accumulated and is therefore captured.

5.2 Boundary map from provided data

Table 1 summarizes the boundary classification for the carbon z-z cells (the cleanest single-polarization axis); the full 128-cell table is in the supplementary log. The boundary is strongly anisotropic in ϕ_B :

- **Parallel** ($\phi_B = 75^\circ$, $\Delta\mathbf{k} \parallel \mathbf{B}$): the theory holds to $B \approx 1000 \text{ T}$ (10 MG), then breaks at $B \approx 1600\text{--}5000 \text{ T}$ ($\eta = 0.33\text{--}0.42$). The residual grows with $|B|$, the signature of the warm-fluid drive upper-bound failure.
- **Intermediate** ($\phi_B = 30^\circ$): the theory holds throughout the swept range (to 64 MG), $\eta \leq 0.18$.
- **Perpendicular** ($\phi_B = -13^\circ$, $\Delta\mathbf{k} \perp \mathbf{B}$): the theory holds only to $B \approx 500 \text{ T}$ (5 MG), then breaks sharply ($\eta = 0.87\text{--}1.15$ at $B \geq 1000 \text{ T}$). Here the PIC gain collapses to near zero while the baseline, which retains only the unmagnetized-ion response, does not; this is the single-IAW (mechanism iii) failure.

Table 1: Primary results table. Vlasov gain g_V and PIC gain g_{PIC} (mean $\pm$ the t_1-t_2/f_y spread) at $\Delta\lambda = 0.9 \text{ \AA}$. η is the symmetric relative residual (Sec. 3); classification per the pre-registered threshold $\eta = 0.30$. Attributed mechanism: (iii) single-IAW mediation, (i) warm-fluid drive, (C3) comparison-validity offset. The full 128-cell table is in the supplementary log; cells with $|g| < 0.2$ or $\sigma_{\text{PIC}} > |g|$ are flagged illustrative.

Sweep cell	($ \mathbf{B} , \phi_B, \text{pol}$)	g_V	g_{PIC}	η	Class	Mechanism
c-zz- $\phi 75$ -B0	(0, 75, zz)	1.12	1.12 ± 0.27	0.00	holds	—
c-zz- $\phi 75$ -B500	(500, 75, zz)	1.04	1.03 ± 0.27	0.01	holds	—
c-zz- $\phi 75$ -B1000	(1000, 75, zz)	0.84	1.03 ± 0.29	0.18	holds	—
c-zz- $\phi 75$ -B5000	(5000, 75, zz)	0.36	0.53 ± 0.36	0.33	breaks	(i) drive
c-zz- $\phi 30$ -B500	(500, 30, zz)	1.02	0.94 ± 0.28	0.07	holds	—
c-zz- $\phi 30$ -B1000	(1000, 30, zz)	0.79	0.72 ± 0.30	0.09	holds	—
c-zz- $\phi 30$ -B5000	(5000, 30, zz)	0.52	0.49 ± 0.20	0.04	holds	—
c-zz- ϕ -13-B500	(500, -13, zz)	1.04	0.73 ± 0.18	0.30	holds	—
c-zz- ϕ -13-B1000	(1000, -13, zz)	0.93	0.13 ± 0.30	0.87	breaks	(iii)
c-zz- ϕ -13-B5000	(5000, -13, zz)	0.50	-0.04 ± 0.26	1.08	breaks	(iii)
c-zy- $\phi 75$ -B1000	(1000, 75, zy)	0.26	-0.14 ± 0.20	1.52	breaks	(C3) offset
c-zy- $\phi 75$ -B5000	(5000, 75, zy)	0.61	0.50 ± 0.15	0.18	holds	—
c-zy- $\phi 30$ -B1000	(1000, 30, zy)	0.27	0.32 ± 0.16	0.17	holds	—
c-zy- $\phi 30$ -B5000	(5000, 30, zy)	0.47	0.47 ± 0.16	0.01	holds	—
c-zy- ϕ -13-B1000	(1000, -13, zy)	0.04	-0.22 ± 0.22	1.18	breaks	(iii)
c-zy- ϕ -13-B5000	(5000, -13, zy)	0.47	-0.10 ± 0.26	1.21	breaks	(iii)
h-zz- $\phi 15$ -B0 (Te 0.4)	(0, 15, zz)	1.90	2.05	0.07	holds	—
h-zz- $\phi 15$ -B0 (Te 1.2)	(0, 15, zz)	0.70	0.67	0.05	holds	—

Compactness summary: $\Delta\mathbf{k}$ -B angle (equivalently ϕ_B) and $|\mathbf{B}|$ carry the boundary signal; polarization and T_e act as $O(1)$ modifiers.

5.3 New reduced-scale PIC runs

Ten new reduced-scale EPOCH runs extend the sweep to a laser-intensity axis and a strong-field point the original study does not cover. All gains are reduced-scale ($L_x = 200\lambda_0$) and are compared to the baseline evaluated at the same L_x .

Intensity axis (mechanism ii / competing processes). At $\phi_B = 30^\circ$, $B = 1000 \text{ T}$, the z-z gain grows sub-linearly with intensity: $g = \{0.310, 0.382, 0.538\}$ at $I = \{0.5, 1.0, 2.0\} \times 10^{15} \text{ W/cm}^2$ (ratios $1.23\times$ and $1.74\times$ versus the linear prediction $2\times$ and $4\times$), so the residual η grows with I ($0.26 \rightarrow 0.55 \rightarrow 0.68$). The z-y gain is near-saturated ($0.241 \rightarrow 0.261$). The sub-linear, saturating intensity scaling is the signature of the linear, ponderomotive-free ion response failing at high intensity (pre-stated H3), though it is not cleanly separable from intensity-dependent competing processes (spontaneous backscatter and noise-seeded instabilities, which the original study reports onset at doubled intensity). These cells are single-seed (flagged illustrative); a three-seed replication of the $I = 2.0$ z-z cell gives $g = \{0.538, 0.571, 0.563, 0.559\}$ (fixed seed plus three random seeds), i.e. a seed-to-seed standard deviation of 0.012, much smaller than the t_1-t_2/f_y spread (≈ 0.21), so the sub-linear intensity scaling is robust to PIC-noise realization.

Strong-field extension (mechanism i). At $B = 8000 \text{ T}$ (80 MG, $\omega_{ce}/\omega_{\text{laser}} = 0.26$), the PIC z-z gain remains $g \approx 0.30$ at both $\phi_B = 30^\circ$ and 75° , whereas the baseline's Faraday-overlap model predicts a stronger suppression ($g_V = 0.15\text{--}0.19$); $\eta = 0.37\text{--}0.51$ (breaks). The z-y cells are well captured ($\eta = 0.05\text{--}0.28$). This is the strong-field boundary where the warm-fluid ponderomotive drive's magnetization correction is no longer negligible (pre-stated H1): the measured z-z coupling persists beyond the cold-plasma overlap prediction.

Validation summary. The reduced-scale reference cell reproduces the baseline within 29% (Sec. 5.1), and the reduced-scale B -trend is flat, consistent with the L_x -attenuated Faraday overlap. The new runs therefore resolve the *local* intensity and strong-field boundaries while the L_x -accumulated Faraday B -dependence is carried by the provided (full-length) data.

6 Mechanism attribution

Each boundary segment is attributed to a pre-registered mechanism hypothesis. The attributions are summarized in Table 1 and detailed below.

Perpendicular boundary → mechanism (iii) (single-IAW mediation). The $\phi_B = -13^\circ$ breakdown is localized where the beat wavevector is nearly perpendicular to $\mathbf{B}$, i.e. where the ion-acoustic mediation along unperturbed gyro-orbits is invalid. The residual is large only where the ion finite-Larmor-radius parameter $k_\perp \rho_i$ is $\gtrsim 1$ (Table 1): at $B = 1000$ T, $k_\perp \rho_i \approx 7.7$ in the perpendicular cell versus ≈ 0 in the parallel cell. This is precisely the regime in which the magnetized-ion susceptibility departs from the unmagnetized form and weakly-damped magnetized modes (ion-Bernstein, lower-hybrid) enter. The attribution is *pre-stated* (H4) and is supported by the correlation of η with $k_\perp \rho_i$ across cells, not by post-hoc selection.

Parallel strong-field boundary → mechanism (i) (warm-fluid drive). The $\phi_B = 75^\circ$ breakdown at $B \gtrsim 1600$ T is the strong-field tail where $\omega_{ce}/\omega_{\text{laser}}$ is no longer $\ll 1$ ($\omega_{ce}/\omega_{\text{laser}} \approx 0.05$ – 0.16 over $B = 1600$ – 5000 T). The warm-fluid ponderomotive drive drops the magnetization correction, which is only justified in the window $\Delta\omega \ll \omega_{ce} \ll \omega_{\text{laser}}$; the upper bound is approached here. The attribution is pre-stated (H1).

Intensity boundary → mechanism (ii) (nonlinear ion response). The new intensity-axis runs (Sec. 5.3) test the linear, ponderomotive-free ion response (pre-stated H3). The measured z-z gain scales sub-linearly with intensity ($0.310 \rightarrow 0.382 \rightarrow 0.538$ over a $4\times$ intensity range) and the residual grows monotonically with I , the predicted signature of the ion nonlinear response becoming non-negligible. Because intensity also activates competing processes (backscatter, noise-seeded instabilities), the attribution is reported as “mechanism (ii) or comparison-validity (C2/C3)” rather than uniquely mechanism (ii); the pre-stated hypothesis is supported by the monotonic residual trend but not cleanly isolated.

Comparison-validity prerequisites. The z-y cells at low B carry a background offset $g_0 \approx -0.2$ to -0.4 (side- and back-scattering) that the offset-free theory does not model; these cells are flagged as comparison-limited rather than theory failures (C3). The hydrogen temperature axis is confounded by species, density, and geometry changes and is interpreted only as a scaling check.

7 Compactness analysis

The boundary is compact. A Pareto-style decomposition of which axes carry the agreement-disagreement signal shows that the $\Delta\mathbf{k}$ – $\mathbf{B}$ angle (equivalently ϕ_B) is the dominant dimension: it moves the boundary from ~ 50 MG (parallel) to ~ 5 MG (perpendicular), a factor of ten. $|\mathbf{B}|$ is the second dimension, setting where along each ϕ_B the boundary is crossed. Polarization is a modifier, not an independent axis: z-z and z-y break at the same ϕ_B values, differing only in the offset and activation threshold. Temperature (the confounded hydrogen axis) changes the gain magnitude and the boundary field strength by $O(1)$ but does not create a new boundary direction. The compact summary is therefore: *the $\Delta\mathbf{k}$ – $\mathbf{B}$ angle and $|\mathbf{B}|$ carry the boundary signal; everything else modifies it by factors of order unity.* We did not find a fully entangled high-dimensional surface; the two dominant axes capture the behavior.

8 Failure analysis

Three aspects of the analysis are genuinely weak and are reported as such.

(i) *Near-zero and offset-dominated cells.* The z-y low- B cells and the perpendicular z-z cells at $B \geq 1000$ T have $|g_{\text{PIC}}| \lesssim 0.3$, near or below the offset and the noise floor. Their “breaks” classification reflects the metric’s floor, not a robustly measured theory failure; they are flagged illustrative in Table 1. (ii) *The magnetized-ion response was not brought to a quantitative baseline.* We implemented the cold-plasma eigenmodes and the unmagnetized IAW response exactly, but the magnetized-ion susceptibility was reduced to a diagnostic ($k_\perp \rho_i$) rather than a full baseline; the mechanism (iii) attribution is therefore correlational, not a closed residual. (iii) *H2 was untested.* The lower-bound drive failure (beat frequency approaching ω_{ce}) requires weak fields or large detunings not present in the provided data; it is reported as untested rather than claimed.

A coherent pattern emerges: the theory is most reliable where a single, unmagnetized, strongly-damped IAW mediates the transfer, and it degrades when the field makes either the mediating wave

or the drive non-single-mode. The failure modes are not spurious; they line up with the two pre-registered physical mechanisms.

9 Computational cost

The Vlasov-baseline solver (eigenmode diagonalization plus the resonant response, all in NumPy/SciPy) evaluates a full 128-cell sweep in seconds on one CPU core. The reduced-scale EPOCH runs take ~ 29 minutes each on 16 vCPU (factor ~ 8 cheaper than the paper’s full-resolution configuration). The cost–coverage tradeoff is favorable: adding five more PIC runs at the reduced scale (~ 2.4 CPU-hours) would extend the intensity axis and the strong-field tail, but would not change the dominant perpendicular/parallel boundary structure, which is already resolved by the provided data. The practical asymmetry is the point of the map: the Vlasov baseline is $\sim 10^5$ times cheaper per point than PIC, so knowing where it is valid has immediate design value.

10 Related work

Magnetized laser–plasma interactions have a long literature. The kinetic theory of three-wave coupling in magnetized plasmas was developed by Stenflo and collaborators [8]; magnetized stimulated Raman scattering was treated kinetically by Boyd and Rankin [9]; and PIC studies have shown external fields modifying Raman growth [10]. The companion Vlasov theory [5] and its PIC validation [6] are the immediate antecedents. Unmagnetized CBET is well understood [11, 12, 13, 14], and the present work does not duplicate any of these methods; it uses the standard linearized kinetic CBET framework [11, 12] as the reconstruction anchor and adds a validity-boundary analysis, which to our knowledge has not been applied to MagCBET. The reduced-scale PIC validation approach (reproduce a reference point at lower cost before sweeping) is standard practice in the EPOCH literature [7].

11 Implications for ICF practice

The map translates directly to design guidance. (1) For near-parallel $\Delta\mathbf{k}$ and $\mathbf{B}$ at $|\mathbf{B}| \lesssim 10$ MG—the geometry of most indirect-drive and magnetized direct-drive beam crossings—the fast Vlasov prediction is adequate and PIC is unnecessary. (2) For near-perpendicular $\Delta\mathbf{k}$ and $\mathbf{B}$ above a few MG, the Vlasov single-IAW prediction is unreliable and full kinetic simulation is required; this is the regime where magnetized ion modes (ion-Bernstein, lower-hybrid) can mediate additional transfer that the single-IAW theory misses. (3) At the highest fields ($\gtrsim 50$ MG), even the parallel geometry requires care, as the drive magnetization correction is no longer negligible. (4) For intensity above $\sim 10^{15}$ W/cm², ion nonlinearities begin to matter (Sec. 5.3). A practitioner can therefore select the theory-versus-PIC choice from a two-dimensional lookup on the $\Delta\mathbf{k}$ – $\mathbf{B}$ angle and $|\mathbf{B}|$, with the intensity as a secondary check.

12 Limitations and conclusions

We have produced a compact, statistically-supported boundary map separating where the linearized Vlasov MagCBET theory remains predictive from where it breaks, in a stated subspace of ($|\mathbf{B}|$, ϕ_B , n_e , T_e , I_{laser} , polarization). The boundary is dominated by the $\Delta\mathbf{k}$ – $\mathbf{B}$ angle: parallel crossings remain predictive to ~ 10 MG, perpendicular crossings break at ~ 5 MG. The perpendicular breakdown is attributed to the failure of the single strongly-damped ion-acoustic mediation assumption (magnetized ion modes), and the strong-field breakdown to the warm-fluid ponderomotive-drive approximation; both attributions were pre-stated. A new intensity boundary was mapped with reduced-scale PIC runs.

The principal limitations are the partial (unmagnetized-ion) baseline, the confounded hydrogen temperature axis, and the noise-floor ambiguity of near-zero-gain cells, all of which are disclosed. These do not change the headline result: for ICF design, the fast Vlasov MagCBET prediction can be trusted in the near-parallel, sub-10-MG regime, and full kinetic simulation is needed elsewhere—most urgently in the near-perpendicular geometry that activates magnetized ion dynamics.

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